



AFCO 8680

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations
Revision Date: 1/25/2018

Version: 1.0

SECTION 1: IDENTIFICATION

Product Identifier

Product Form: Mixture

Product Name: AFCO 8680

Product Code: AFCO 8680

Intended Use of the Product

Use of the Substance/Mixture: A microbiocidal bactericide, fungicide, algicide, and slimicide. For professional use only.

Name, Address, and Telephone of the Responsible Party

Company

Alex C. Fergusson, LLC.
800 Development Avenue
Chambersburg, PA 17201
T: 800-345-1329

www.afcocare.com

Emergency Telephone Number

Emergency Number : 1-800-424-9300 (CHEMTREC)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the Substance or Mixture

Classification (GHS-US)

Acute Tox. 4 H302

Skin Sens. 1A H317

Eye Dam. 1 H318

Label Elements

GHS-US Labeling

Hazard Pictograms (GHS-US)



Signal Word (GHS-US)

: Danger.

Hazard Statements (GHS-US)

: H302 - Harmful if swallowed.
H317 - May cause an allergic skin reaction.
H318 - Causes serious eye damage.

Precautionary Statements (GHS-US)

: P261 - Avoid breathing dust/fume/gas/mist/vapors/spray.
P264 - Wash hands, forearms, and other exposed areas thoroughly after handling.
P270 - Do not eat, drink or smoke when using this product.
P272 - Contaminated work clothing should not be allowed out of the workplace.
P280 - Wear protective glove, protective clothing, eye protection, face protection.
P301+P330+P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting.
P302+352 - IF ON SKIN: Wash with plenty of soap and water.
P303+P361+P353 - IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
P304+P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
P305+P351+P338 - If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310 - Immediately call a POISON CENTER or doctor/physician.
P362+364 - Take off contaminated clothing and wash it before reuse.
P333+313 - If skin irritation or rash occurs, seek medical advice/attention.
P501 - Dispose of contents/container to local, regional, national, territorial, provincial, and international regulations.

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Other Hazards

Other Hazards Not Contributing to the Classification: Exposure may aggravate those with pre-existing eye, skin, or respiratory conditions.

Unknown Acute Toxicity (GHS-US): Not available.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Substances

Mixture

Name	Product identifier	% (w/w)	Classification (GHS-US)
Water	(CAS No) 7732-18-5	30.0	Not classified.
Polyethylene glycol	(CAS No) 25322-68-3	50.0	Not classified
2,2-Dibromo-3-nitrilopropionamide	(CAS No) 10222-01-2	19.0-21.0	Skin Corr. 1B, H314 Skin Sens. 1, H317 Eye Dam. 1, H318 Aquatic Acute 1, H400 Aquatic Chronic 1, H410

Full text of H-phrases: see section 16.

SECTION 4: FIRST AID MEASURES

Description of First Aid Measures

General: Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible).

Inhalation: Remove from exposure and move to fresh air immediately. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical advice.

Skin Contact: Flush skin with plenty of water for several minutes while removing contaminated clothing and shoes. Get medical advice.

Eye Contact: Drench with plenty of water for several minutes, occasionally lifting the upper and lower eyelids. Get medical advice.

Ingestion: Rinse mouth with water. Do NOT induce vomiting. Get medical advice.

Most Important Symptoms and Effects Both Acute and Delayed

General: Causes serious eye damage. Harmful if swallowed. May cause an allergic skin reaction.

Inhalation: May cause respiratory tract irritation.

Skin Contact: May cause sensitization. May cause skin irritation.

Eye Contact: Causes serious eye damage

Ingestion: Harmful if swallowed. May cause irritation of the digestive tract.

Chronic Symptoms: Not available.

Indication of Any Immediate Medical Attention and Special Treatment Needed

If exposed or concerned, get medical advice and attention.

SECTION 5: FIRE-FIGHTING MEASURES

Extinguishing Media

Suitable Extinguishing Media: Water spray, carbon dioxide, dry chemical.

Unsuitable Extinguishing Media: No data available.

Special Hazards Arising From the Substance or Mixture

Fire Hazard: No data available.

Explosion Hazard: No data available.

Reactivity: When heated to decomposition may emit: nitrogen oxides (NO_x), hydrogen bromide, bromine fumes.

Advice for Firefighters

Precautionary Measures Fire: Exercise caution when fighting any chemical fire.

Firefighting Instructions: As in any fire, wear a self-contained breathing apparatus in pressure-demand, MSHA/NIOSH (approved or equivalent), and full protective gear. Use water spray to cool unopened containers.

Protection During Firefighting: Do not enter fire area without proper protective equipment, including respiratory protection.

Hazardous Combustion Products: Nitrogen oxides (NO_x). Hydrogen bromide. Bromine fumes.

Other information: Do not allow run-off from fire fighting to enter drains or water courses.

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Reference to Other Sections

Refer to section 9 for flammability properties.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

General Measures: Avoid release to the environment. Avoid breathing (vapors, mist, spray). Avoid all contact with skin, eyes, or clothing.

For Non-Emergency Personnel

Protective Equipment: Use appropriate personal protection equipment (PPE) as indicated in Section 8.

Emergency Procedures: Evacuate unnecessary personnel.

For Emergency Personnel

Protective Equipment: Equip cleanup crew with proper protection.

Emergency Procedures: Ventilate area.

Environmental Precautions

Prevent entry to sewers and public waters. Avoid release to the environment.

Methods and Material for Containment and Cleaning Up

For Containment: Decontaminate spill area with 10% sodium bicarbonate solution. Absorb spill with inert material (e.g. vermiculite, sand or earth), then place in suitable container.

Methods for Cleaning Up: Clear up spills immediately and dispose of waste safely. Contact competent authorities after a spill.

Reference to Other Sections

See heading 8, Exposure Controls and Personal Protection.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Additional Hazards When Processed: When heated to decomposition may emit: nitrogen oxides (NO_x), hydrogen bromide, bromine fumes.

Hygiene Measures: Handle in accordance with good industrial hygiene and safety procedures. Wash hands and other exposed areas with mild soap and water before eating, drinking, or smoking and again when leaving work. Do not eat, drink or smoke when using this product. Wash hands and forearms thoroughly after handling.

Conditions for Safe Storage, Including Any Incompatibilities

Technical Measures: Comply with applicable regulations.

Storage Conditions: Store in a dry, cool and well-ventilated place. Keep container tightly closed when not in use. Keep/Store away from extremely high or low temperatures, incompatible materials.

Incompatible Materials: Strong oxidizing agents. Reducing agents. Bases.

Special Rules on Packaging: Keep only in original container.

Specific End Use(s) A microbiocidal bactericide, fungicide, algicide, and slimicide. For professional use only.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

No Occupational Exposure Limits (OELs) have been established for this product or any of its chemical components

Exposure Controls

Appropriate Engineering Controls: Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. Ensure all national/local regulations are observed. Ensure adequate ventilation, especially in confined areas. If user operations generate fumes, vapor, spray, or mist use process enclosures, local exhaust ventilation, or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits.

Personal Protective Equipment: Splash goggles and/or face shield. Lab coat and/or apron. Impervious gloves.



Materials for Protective Clothing: Chemically resistant materials and fabrics.

Hand Protection: Impervious gloves.

Eye Protection: Splash goggles and/or face shield.

Skin and Body Protection: Lab coat and/or apron.

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Respiratory Protection: Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Other Information: When using, do not eat, drink or smoke.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on Basic Physical and Chemical Properties

Physical State	: Liquid.
Appearance	: Clear, yellow to amber.
Odor	: Chlorine-like.
Odor Threshold	: Not available.
pH	: 2.8 (10% soln)
Relative Evaporation Rate (butylacetate=1)	: Not available.
Melting Point	: Not available.
Freezing Point	: <30°F (<-34.4°C)
Boiling Point	: >70°C (>158°F)
Flash Point	: Not available.
Auto-ignition Temperature	: Not available.
Decomposition Temperature	: >70°C (>158°F)
Flammability (solid, gas)	: Not available.
Lower Flammable Limit	: Not available.
Upper Flammable Limit	: Not available.
Vapor Pressure	: 4x10 ⁻⁵ MM_HG at 25°C (77°F)
Relative Vapor Density at 20°C	: <1
Specific Gravity	: 1.25
Solubility	: Soluble in water.
Partition coefficient: n-octanol/water	: Not available.
Viscosity	: Not available.
Explosion Data – Sensitivity to Mechanical Impact	: Not expected to present an explosion hazard due to mechanical impact.
Explosion Data – Sensitivity to Static Discharge	: Not expected to present an explosion hazard due to static discharge.

SECTION 10: STABILITY AND REACTIVITY

Reactivity: When heated to decomposition may emit: nitrogen oxides (NO_x), hydrogen bromide, bromine fumes.

Chemical Stability: Stable when used as directed.

Possibility of Hazardous Reactions: Hazardous polymerization will not occur.

Conditions to Avoid: Direct sunlight. Extremely high or low temperatures. Heat. Incompatible materials.

Incompatible Materials: Strong oxidizing agents. Reducing agents. Bases.

Hazardous Decomposition Products: Nitrogen oxides (NO_x). Hydrogen bromide. Bromine fumes.

SECTION 11: TOXICOLOGICAL INFORMATION

Information on Toxicological Effects - Product

Acute Toxicity: Not classified.

LD50 and LC50 Data: Not available.

Skin Corrosion/Irritation: May cause sensitization or skin irritation. **pH(10%):** 2.8

Serious Eye Damage/Irritation: Causes serious eye damage. **pH(10%):** 2.8

Respiratory or Skin Sensitization: Not classified.

Germ Cell Mutagenicity: Not classified.

Teratogenicity: Not available.

Carcinogenicity: Not classified.

Specific Target Organ Toxicity (Repeated Exposure): Not classified.

Reproductive Toxicity: Not classified.

Specific Target Organ Toxicity (Single Exposure): Not classified.

Aspiration Hazard: Not classified.

Symptoms/Injuries After Inhalation: May cause respiratory tract irritation.

Symptoms/Injuries After Skin Contact: May cause sensitization. May cause skin irritation.

Symptoms/Injuries After Eye Contact: Causes serious eye damage.

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Symptoms/Injuries After Ingestion: Harmful if swallowed. May cause irritation of the digestive tract.

Information on Toxicological Effects - Ingredient(s)

LD50 and LC50 Data: Not available.

SECTION 12: ECOLOGICAL INFORMATION

Toxicity Not available.

2,2-Dibromo-3-nitrilopropionamide (10222-01-2)	
LC50 Fish 1	2.3 mg/L (Exposure time: 96 h - Species: Lepomis macrochirus [Bluegill])
LC50 Fish 1	1.0 mg/l (Exposure time: 96 h - Species: Oncorhynchus mykiss)
EC50 Daphnia 1	0.86 mg/l (Exposure time: 48 h - Species: Daphnia magna)
Polyethylene glycol (25322-68-3)	
LC50 Fish	>100 mg/l (Exposure time: 96h - Species: Leuciscus idus)

Persistence and Degradability

AFCO 8680	
Persistence and Degradability	Not established.

Bioaccumulative Potential

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Bioaccumulative Potential	Not established.

Mobility in Soil Not available.

Other Adverse Effects

Other Information: Avoid release to the environment.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Recommendations: Biocide or Pesticide wastes are acutely hazardous. Improper disposal or excess product or rinsate is a violation of Federal law. If these wastes cannot be disposed of by use according to label instructions, contact your State Biocide/Pesticide or Environmental Control Agency, or the Hazardous Waste representative at your nearest EPA Regulation Office for guidance.

Ecology - Waste Materials: RCRA: It is the responsibility of the product user to determine at the time of disposal whether a material containing the product or derived from the product should be classified as a hazardous waste (40 CFR 261.20-24).

SECTION 14: TRANSPORT INFORMATION

14.1 In Accordance with DOT	:	Not regulated for transport.
14.2 In Accordance with IMDG	:	Not regulated for transport.
14.3 In Accordance with IATA	:	Not regulated for transport.
14.4 In Accordance with TDG	:	Not regulated for transport.

SECTION 15: REGULATORY INFORMATION

US Federal Regulations

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SARA Section 311/312 Hazard Classes	Immediate (Acute) health hazard. Delayed (Chronic) health hazard. Sudden Release of Pressure Hazard.
Water (7732-18-5)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
2,2-Dibromo-3-nitrilopropionamide (10222-01-2)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Polyethylene glycol (25322-68-3)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	

US State Regulations

2,2-Dibromo-3-nitrilopropionamide (10222-02-1)	
RTK - U.S. - Massachusetts - Right To Know List.	
RTK - U.S. - New Jersey - Right to Know Hazardous Substance List.	
RTK - U.S. - Pennsylvania - RTK (Right to Know) List.	

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Canadian Regulations

Water (7732-18-5)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Polyethylene glycol (25322-68-3)

Listed on the Canadian DSL (Domestic Substances List) inventory.

SECTION 16: OTHER INFORMATION, INCLUDING DATE OF PREPARATION OR LAST REVISION

Revision Date : 1/25/2018

Other Information : This document has been prepared in accordance with the SDS requirements of the OSHA Hazard Communication Standard 29 CFR 1910.1200.

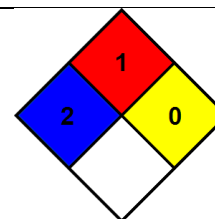
GHS Full Text Phrases:

Acute Tox. 4 (Oral)	Acute toxicity (oral) Category 4.
Aquatic Acute 1	Hazardous to the aquatic environment - Acute Hazard Category 1.
Eye Dam. 1	Serious eye damage/eye irritation Category 1.
Eye Irrit. 2A	Serious eye damage/eye irritation Category 2A.
Eye Irrit. 2B	Serious eye damage/eye irritation Category 2B.
Skin Irrit. 2	Skin corrosion/irritation Category 2.
Skin Irrit. 3	Skin corrosion/irritation Category 3.
STOT SE 3	Specific target organ toxicity single exposure Category 3.
H302	Harmful if swallowed.
H317	May cause an allergic skin reaction.
H318	Causes serious eye damage.
H319	Causes serious eye irritation.
H320	Causes eye irritation.
H335	May cause respiratory irritation.
H400	harmful to aquatic life with long lasting effects.

NFPA Health Hazard : 2 - Intense or continued exposure could cause temporary incapacitation or possible residual injury unless prompt medical attention is given.

NFPA Fire Hazard : 1 - Must be preheated before ignition can occur.

NFPA Reactivity : 0 - Normally stable, even under fire exposure conditions, and are not reactive with water.



HMIS III Rating

Health : 2 Moderate Hazard - Temporary or minor injury may occur.

Flammability : 1 - Slight Hazard.

Physical : 0 - Minimal Hazard.

Party Responsible for the Preparation of This Document

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This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

NA GHS SDS 2015 (U.S., Can., Mex.)